

Poisson distribution $\rightarrow \frac{\lambda^0}{0!} + \frac{\lambda^1}{1!} + \frac{\lambda^2}{2!} + \dots$

$$P(X=i) = \frac{n!}{i!(n-i)!} \left(\frac{\lambda}{n}\right)^i \left(1 - \frac{\lambda}{n}\right)^{n-i}$$

$$\lim_{n \rightarrow \infty} P(X=i) = \frac{n(n-1)\dots(n-i+1)}{i!} \frac{\lambda^i}{n^i} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-i}$$

$$= \frac{\lambda^i}{i!} \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n = \frac{e^{-\lambda} \lambda^i}{i!}$$

Poisson PMF

$$\lim_{n \rightarrow \infty} n \log \left(1 - \frac{\lambda}{n}\right) = \lim_{n \rightarrow \infty} \frac{\log \left(1 - \frac{\lambda}{n}\right)}{\frac{-1/n}{1 - \lambda/n}}$$

$$= \lim_{n \rightarrow \infty} \frac{\frac{-\lambda/n}{1 - \lambda/n}}{-1/n} = -\lambda$$